

Nicholas Drazso

[Design Portfolio](#)

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| 905-244-1804

SKILLS

Mechanical: CATIA 3DX, SolidWorks, DFM, Sheet Metal Design, 3D Printing, Machining, Welding

Electrical: Altium, Wire Harness Design, Soldering, Arduino, RPi

Software: C++, Python, MATLAB, Simulink, Linux, LTSpice, PSIM

EXPERIENCE

Co – Founder | *LIDDE Loupes* | *Waterloo, ON, CA* *Sept 2025 – Present*

- Technical lead driving system architecture for a medical real time vision wearable
- Designing, integrating, and validating mechanical, high-speed PCB, vision, and power subsystems
- Secured \$25k in non-dilutive funding and 20+ LOI's from surgeons and dentists

HV Battery Test Development Engineer | *Tesla* | *Palo Alto, CA, USA* *May 2025 – Aug 2025*

- Led a DoE to validate a \$10k manual HV bolted joint test method for production, avoiding a \$500k automated system, improving yield by 83%, and defining a test standard
- Reduced false pass and escape risk in leak testing by developing custom hardware and software tools
- Optimized testing from cell to pack level through simulation and hands on validation including electrical, thermal, and leak testing

Mechatronics Design | *Electrans* | *Oakville, ON, CA* *Sept 2024 – Dec 2024*

- Derisked chance of \$200k driver related damages by implementing an actively controlled lift system
- Enabled reliable PCB testing at scale by creating a bed of nails test fixture with custom PCB and RPi
- Reduced custom connector cost by 80% by optimizing PCBA layout and over molding volume

Cell R&D Test Equipment Design | *Tesla* | *Palo Alto, CA, USA* *Jan 2024 – Apr 2024*

- Enabled traceability of 10k+ PCBAs by designing an EEPROM programming fixture utilizing Pytest
- Launched a new novel cell test fixture by root causing failures and implementing 6 design changes
- Cut thermocouple data visualization time by 50% by developing a custom visualization app
- Improved efficiency and accuracy of novel cylindrical, prismatic, coin, and pouch cell test fixtures

Mechatronics Design | *BotBuilt* | *Durham, NC, USA* *Sept 2022 – Dec 2022*

- Increased system reliability by designing an autonomous unit test fixture for a novel end effector
- Improved scalability of manufacturing by implementing 5 critical design changes to end effectors

EXTRA CURRICULARS

Autonomous Search and Rescue Robot | *Third Year Design Project* *Jan 2025 – Mar 2025*

- Created a 4-layer PCB chassis integrating motor drivers, power regulation, and a Teensy 4.0 MCU
- Designed a suction fan system to increase downforce resulting in a 5% cornering speed increase
- Placed 1st out of 17 teams, scoring 68x higher than 2nd place based on speed, weight, and reliability
- Tuned encoder-based PD control with Pixy 2.1 vision module for line tracking enabling 13s run

Mechanical Team Lead | *UW DeepBlue* | *Waterloo, ON, CA* *Sept 2024 – Dec 2024*

- Led the design of 4 electromechanical subsystems and hull for an underwater autonomous vehicle

EDUCATION

BASc Mechatronics Engineering | *The University of Waterloo*

Sept 2019 – Apr 2026